
```

[3] great_ne(X1, X2) :- x_subst(X2, X3) and r_subst_1(X1, X3).
    Training Acc: 0.7689 | Eval Acc: 0.7609

[4] great_ne(X1, X2) :- gt(X1, X3) and great_pi_acc(X2, X3).
    Training Acc: 0.7689 | Eval Acc: 0.7609

[5] great_ne(X1, X2) :- not(ring_subst_4(X2, X3)) and ring_subst_4(X3, X1).
    Training Acc: 0.7689 | Eval Acc: 0.7609

[6] great_ne(X1, X2) :- great_ne(X3, X2) and flex(X1, X3).
    Training Acc: 0.7689 | Eval Acc: 0.7609

[7] great_ne(X1, X2) :- r_subst_2(X3, X2) and ring_subst_2(X3, X1).
    Training Acc: 0.7689 | Eval Acc: 0.7609

[8] great_ne(X1, X2) :- great_ne(X3, X2) and ring_subst_2(X3, X1).
    Training Acc: 0.7689 | Eval Acc: 0.7609

[9] great_ne(X1, X2) :- pi_doner(X3, X1) and not(ring_substitutions(X2, X3)).
    Training Acc: 0.7689 | Eval Acc: 0.7609

[10] great_ne(X1, X2) :- great_ne(X2, X3) and pi_doner(X3, X1).
    Training Acc: 0.7689 | Eval Acc: 0.7609

```

K.2 UW-CSE DATASET

The *UW-CSE* dataset models academic relationships within a university domain, including roles, courses, projects, and advising relationships. The task is to infer latent advisory relations from heterogeneous academic facts.

$\text{ADVISED BY}(X_1, X_2)$

The target predicate $\text{advisedby}(X_1, X_2)$ represents an academic advising relationship between individuals. Below, we present representative rules extracted by ANDRE, highlighting recursive, role-based, and relational patterns commonly observed in academic environments.

Listing 9: Extracted Rules for $\text{advisedby}(X_1, X_2)$ (UW-CSE Dataset)

```

[1] advisedby(X1, X2) :- advisedby(X3, X1) and yearsinprogram(X3, X2).

[2] advisedby(X1, X2) :- courselevel(X1, X3) and hasposition(X2, X3).

[3] advisedby(X1, X2) :- professor(X3, X2) and taughtby(X3, X1).

[4] advisedby(X1, X2) :- courselevel(X2, X3) and not(professor(X3, X1)).

[5] advisedby(X1, X2) :- advisedby(X3, X1) and not(inphase(X3, X2)).

[6] advisedby(X1, X2) :- hasposition(X2, X3) and ta(X3, X1).

[7] advisedby(X1, X2) :- not(hasposition(X3, X1)) and projectmember(X2, X3).

[8] advisedby(X1, X2) :- inphase(X1, X3) and professor(X2, X3).

[9] advisedby(X1, X2) :- not(courselevel(X3, X2)) and publication(X1, X3).

[10] advisedby(X1, X2) :- hasposition(X2, X3) and inphase(X3, X1).

```

L SYNTHETIC DATASETS TABULAR RESULTS

Table 9: Comparison of Rule Extraction Performance between ANDRE and DFORL on Complex Synthetic Datasets with Varying Number of Subrules

Dataset	Sample Size	Accuracy				Rule Extraction Success	
		ANDRE		DFORL		ANDRE	DFORL
		Train	Test	Train	Test		
R ₁	20	0.95	0.80	0.85	0.84	✗	✗
	50	0.98	0.96	0.82	0.84	✓	✗
	100	0.95	0.96	0.83	0.85	✓	✓
	200	0.96	0.96	0.86	0.84	✓	✓
R ₂	50	0.94	0.88	0.78	0.74	✗	✗
	100	0.95	0.96	0.74	0.73	✓	✗
	200	0.95	0.95	0.73	0.73	✓	✗
	500	0.95	0.95	0.75	0.73	✓	✗
	1000	0.96	0.97	0.74	0.72	✓	✗
R ₃	200	0.87	0.83	0.75	0.79	✗	✗
	500	0.95	0.95	0.68	0.68	✓	✗
	1000	0.90	0.90	0.68	0.70	✓	✗
	1500	0.90	0.90	0.68	0.67	✓	✗
	2000	0.90	0.89	0.68	0.65	✓	✗

Table 10: Comparison of Rule Extraction Performance between ANDRE and DFORL on Noisy Synthetic Datasets

Dataset	Sample Size	Noise (%)	Accuracy				Rule Extraction Success	
			ANDRE		DFORL		ANDRE	DFORL
			Train	Test	Train	Test		
R ₄	200	10	0.86	0.86	0.71	0.66	✓	✓
		20	0.79	0.74	0.68	0.61	✓	✗
		25	0.74	0.71	0.65	0.60	✓	✗
		30	0.69	0.61	0.64	0.58	✗	✗
R ₅	500	5	0.90	0.90	0.62	0.56	✓	✗
		15	0.82	0.82	0.58	0.54	✓	✗
		25	0.73	0.72	0.56	0.54	✓	✗
		35	0.62	0.55	0.54	0.51	✗	✗
		45	0.62	0.55	0.49	0.47	✗	✗
R ₆	1000	5	0.84	0.82	0.49	0.49	✓	✗
		10	0.80	0.78	0.49	0.48	✓	✗
		15	0.76	0.76	0.49	0.48	✓	✗
		20	0.73	0.71	0.48	0.48	✗	✗
		25	0.70	0.67	0.49	0.48	✗	✗